



Patent pending

either put in together or buy it prebuilt and with the soon to be released Filabot Reclaimer, you can convert old projects in to filaments too!

Phew! Now that that's out of the way, let's talk about dispensing the filament. A lot of people have come up with a lot of clever, inventive ways to dispense the filament but an easy way developed by the author (patent pending by the way) would be to make one with a few

threaded rods. Imagine two triangles joined by more rods (very similar to the frame of our printer). The spool of filament would be threaded through the connecting rods and get dispensed by the extruder.

Programmes

To actually build your model, you will need three programmes at the most. One will be a CAD programme to actually make your model, one will convert the model in to G-code and the third will communicate with the printer and produce the model. CAD programmes are numerous. While you can pick the one that suits your needs best from reprap.org/wiki/Useful_Software_Packages#2D_and_3D_CAD_software, a few open source ones worth talking about would be Blender, FreeCAD and Shapsmith.

Blender is a really powerful open source 3D animation suite which also allows the user to sculpt objects. It is entirely community driven which has led to a lot of bug fixes and there are also a whole bunch of training DVDs you can purchase and help support Blender.

FreeCAD, like Blender works across multiple operating systems and is more suited for 3D modelling. Its official website freecadweb.org has all the resources you would need to get started.

Finally, Shapsmith is a browser based programme and because your set parameters to compose your 3D object, it takes less time to make the objects. The only down side is the fact that you can't exactly sculpt your objects here.

Other free but closed source applications you can look at are SketchUp which is supremely easy and super intuitive to use and Rhino which is favoured by the jewellery designing community. The only caveat with Rhino is the fact that you can save a total of 25 times before the save feature is disabled and you will need to purchase a license.